A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, IV: Wild local correspondence and global trace formula comparison

Draft 3

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Abstract

We continue Parts I–III for an odd-degree covering of the multiplicative inner form of GL_r with r prime and $(r, n) = 1$. First we reconcile the split-diagonal cocycle with the ambient-cover restriction and central relabeling in Part I. The regular good-element criterion for *every* maximal torus, including nonsplit tori and wild residue characteristic, is already a theorem of Flicker–Kazhdan; we supply its proof with the parameters used in this series. We prove two refinements useful in global comparison: an almost-everywhere local-to-global criterion for globally good regular classes, and an exact local calculation of the kernel, index, and Haar modulus of the nth-power map on an arbitrary finite *étale* torus modulo scalars. The power-map calculation gives the split-place root multiplicities, while a separate regularity example explains why a regular root need not have a regular nth power. The genuine character transform of Part III is then placed alongside the corrected metaplectic character relation of Zou and the older work of Flicker–Kazhdan. A normalization ledger and explicit Flicker–Kazhdan-compatible split-place test-function conditions separate the previously established covered-inner-form trace comparison from the additional proposed comparison with ordinary $D^\times$. The wild irreducible-character integrality and the latter global comparison are not established here.

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1 Introduction and mathematical status

The problem concerns three different local representation theories. Write $H_v = D_v^\times$, and let $\overline{G_{D_v}}$ denote its specified n-fold central cover; the cover $\overline{G}_v$ of $\mathrm{GL}_r(K_v)$ is defined inside the same ambient cover. An additional *ordinary* group H_v enters through the transform $L_{D,v}$ of Part III. These are not interchangeable. In particular, a comparison between $\overline{G_D}$ and $\overline{G}$ has already been obtained in Parts I–II under its stated test-function and spectral hypotheses. The new proposed comparison between H and $\overline{G_D}$ is a different assertion.

Global assumptions. Let K be a totally imaginary number field with $\mu_n \subset K$, $n \geq 1$ odd. Let $r \geq 2$ be prime, $(r, n) = 1$, let L/K be cyclic of degree r, and let D be a central division K-algebra of degree r containing L. Put $S = \{v : D_v \text{ is division}\}$. Fix integers r', c such that

$$rr' \equiv 1 \pmod{n}, \qquad r - 1 + 2c \equiv 0 \pmod{n}, \qquad c_v \equiv r'c \pmod{n}. \tag{1}$$

Thus $r - 1 + 2rc_v \equiv 0 \pmod{n}$. In fact c_v is independent of v with these choices; the subscript recalls Part I's notation. Locally $F = K_v$ is a nonarchimedean field containing μ_n; p denotes its residue characteristic. We call v *wild* if $p \mid n$.

Status convention. **[P]** denotes a proof included here (possibly invoking **[E]** inputs established in cited publications or earlier parts); **[O]** denotes a proof outlined but not fully established, and **[C]** denotes a conjectural goal requiring new mathematical input. When a proved proposition has explicit unresolved hypotheses, we mark its formal implication [P] and mark the outstanding hypotheses [C] separately, as in Parts I–III.

Assertion	Precise basis	Status
Diagonal and nonsplit good-element criterion, including wild p	[FK, §3]; Part I, §3	[E]
Compatibility of the Part I cocycle with σ_{c_v}	Part I, Prop. 2.3(b); direct calculation	[P]
Power-map kernel, index, and Haar modulus on $T_E(F)/F^\times$	Section 4	[P]
Global good-regular class criterion	Section 5; Part III Hasse principle	[P]
Division-side unitary character transform and tame integral case	Part III, Thms. 4.3, 7.6	[E]
Wild irreducible-character integrality	Section 6	[C]
Covered D–covered GL global simple trace comparison	Parts I–II under their hypotheses	[E]
Covered D–ordinary D global comparison	Section 8	[C]
Removal of auxiliary support and packet hypotheses	Section 9	[C]

2 The exact cocycle convention and star map

This section fixes the one place where a parameter mismatch could invalidate every subsequent constant. The ambient construction is a restriction of a cocycle over L_v, whereas [FK] uses the standard $\mathrm{GL}_r(F)$ Kazhdan–Patterson convention.

2.1 Restriction and relabeling

Let σ_F be the parameter-zero Kazhdan–Patterson cocycle with restriction to the diagonal torus

$$\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j} (a_i, b_j)_{n,F}. \tag{2}$$

The GL subgroup of the ambient cover in Part I has, *as an extension*, cocycle model

$$\sigma_F(g,h)^r \, (\det g, \det h)_{n,F}^c \quad \text{before relabeling } \mu_n. \tag{3}$$

This assertion is Part I, Proposition 2.3(b). Applying the kernel relabeling $\zeta \mapsto \zeta^{r'}$ yields the model

$$\sigma_{c_v}(g,h) = \sigma_F(g,h)(\det g, \det h)_{n,F}^{c_v}. \tag{4}$$

The two parameters c and c_v must be kept separate: c occurs before relabeling and $c_v = r'c \pmod n$ afterward. The isomorphism in (3) is an isomorphism of extensions, not a claim that arbitrary choices of cocycle sections give literally equal functions on all of $\mathrm{GL}_r(F)^2$.

Proposition 2.1 (**[P]** Diagonal commutator audit)**.** *For $t = \mathrm{diag}(t_1, \dots, t_r)$ and $u = \mathrm{diag}(u_1, \dots, u_r)$ in $\mathrm{GL}_r(F)$,*

$$[\widetilde{t}, \widetilde{u}] = \prod_{i,j} (t_i, u_j)_{n,F}^{A_{ij}}, \qquad A = -I_r + (1 + 2c_v)J_r. \tag{5}$$

Here J_r is the all-ones matrix. Its row sums are $r - 1 + 2rc_v \equiv 0 \pmod n$, so scalar lifts commute with the lifted split torus. The radical on $(F^\times/F^{\times n})^r$ consists precisely of diagonal classes.

Proof. The commutator is the ratio $\sigma_{c_v}(t,u)/\sigma_{c_v}(u,t)$. The Hilbert symbol satisfies $(a,b)_n = (b,a)_n^{-1}$; this is *antisymmetry*, valid for all n. The determinant twist contributes $2c_v$ to each entry of A, and the parameter-zero factor contributes 1 off the diagonal. This gives (5). The Hilbert pairing on $X = F^\times/F^{\times n}$ is perfect. As X is killed by n and $(r,n) = 1$, the decomposition $X^r = \Delta X \oplus \{(x_i) : \sum x_i = 0\}$ holds by averaging with r^{-1} modulo n. On these summands A acts, respectively, by $r - 1 + 2rc_v = 0$ and -1. Therefore $\ker A = \Delta X$. □

Remark 2.2. The assumption $(r, n) = 1$ also follows algebraically from $n \mid r - 1 + 2rc_v$, because any common divisor of r and n must divide 1. We retain it because it is a global hypothesis of the series. The oddness of n is used for the ordinary star-map convention below, not for Hilbert-symbol antisymmetry.

2.2 Star and central characters

For an element h of the ordinary group, choose any lift $\widetilde{h}$ and define $h^* = \widetilde{h}^{\,n}$. It is independent of the lift because μ_n is central. In the present scalar-central cover, for $a \in F^\times$,

$$(ah)^* = a^* h^*, \qquad a^* = \widetilde{a}^{\,n}. \tag{6}$$

If h_1, h_2 commute in an ordinary torus, their lifted commutator is in μ_n, and the class-two power identity

$$(\widetilde{h}_1\widetilde{h}_2)^n = \widetilde{h}_1^{\,n}\widetilde{h}_2^{\,n}[\widetilde{h}_2, \widetilde{h}_1]^{n(n-1)/2} = \widetilde{h}_1^{\,n}\widetilde{h}_2^{\,n} \tag{7}$$

uses n odd. Thus star is multiplicative on the lifted torus. This agrees with the convention in [FK, §4] and [Zou, §3]: Zou writes $x^* = u(x)s(x)^n$, where $u(x) = \pm 1$ is the correction defined in [2, §4] and [3, §3.1]. This correction takes values in the central kernel μ_n, so for odd n it belongs to $\{\pm 1\} \cap \mu_n = \{1\}$. Thus $u(x) = 1$; this uses the range of the correction, not merely the multiplicativity identity in Zou's Lemma 3.5. For a scalar $\zeta \in \mu_n(F)$, $\zeta^* = 1$ in the standard diagonal cocycle model: the self-symbol $(\zeta, \zeta)_n = 1$ for odd n, so $s(\zeta I_r)^n = s(I_r) = 1$. This last statement is also invariant under the extension isomorphisms in Part I. For representation characters we first fix a genuine unitary character $\widetilde{\theta}$ of the central preimage $\widetilde{Z}$ and *then* put $\theta(a) = \widetilde{\theta}(a^*)$. The reverse implication is not unique: θ alone does not specify $\widetilde{\theta}$. For $\zeta \in \mu_n(F)$ we have $\zeta^* = 1$, hence $\theta(\zeta) = 1$. Globally the rational splitting gives $a^* = \iota(a^n)$ for $a \in K^\times$; if $\widetilde{\theta}$ is automorphic, it is trivial on the rational splitting, so $\theta(a) = 1$ on $K^\times$. Test functions use the inverse central characters θ^{-1} and $\widetilde{\theta}^{-1}$.

3 Good elements for every maximal torus

A regular maximal torus in $\mathrm{GL}_r(F)$ may be nonsplit even at a place where $D_v \simeq M_r(F)$. This is the situation not treated by the diagonal matrix alone. The following calculation is not new: it is the proof of [FK, §3, Proposition] in the notation of this series, and also underlies Part I, Proposition 3.5.

Theorem 3.1 ([**E**] Flicker–Kazhdan good-element theorem)**.** *Let $E = \prod_{i=1}^k E_i$ be a finite* étale *F-algebra of dimension r, embedded as a regular maximal torus $T_E(F) = E^\times$ in $\mathrm{GL}_r(F)$. Under (1), for any $t \in T_E(F)$ and any lift $\widetilde{t}$,*

$$[\widetilde{t}, \widetilde{u}] = (\mathrm{N}_{E/F}\, t, \mathrm{N}_{E/F}\, u)_{n,F}^{1+2c_v} \prod_{i=1}^{k} (t_i, u_i)_{n,E_i}^{-1}. \tag{8}$$

The radical of this commutator pairing is

$$\{t \in T_E(F) : [\widetilde{t}, \widetilde{u}] = 1 \ \forall u \in T_E(F)\} = F^\times T_E(F)^n. \tag{9}$$

For regular t, this is exactly the criterion that $\widetilde{t}$ is good. Both identities are valid when $p \mid n$.

Proof from the cited formula, included for clarity. The commutator formula is [FK, §3], using [KP, Prop. 0.1.5]; its restriction to a single field is exactly Part I, Proposition 3.1 after relabeling [8]. Write $N = \mathrm{N}_{E/F}$. Norm compatibility of local Hilbert symbols gives

$$(Nt, Nu)_{n,F} = \prod_i (Nt, u_i)_{n,E_i}.$$

Thus (8) equals

$$\prod_i \left((Nt)^{1+2c_v} t_i^{-1}, u_i\right)_{n,E_i}.$$

The nondegeneracy of the nth Hilbert symbol over each E_i (including wildly ramified E_i) implies that the commutator is trivial for all u exactly when

$$t_i \equiv (Nt)^{1+2c_v} \pmod{E_i^{\times n}} \quad (1 \le i \le k). \tag{10}$$

This directly says $t \in F^\times E^{\times n}$. Conversely, if $t = ay^n$ for $a \in F^\times$, then $Nt = a^r (Ny)^n$; therefore $(Nt)^{1+2c_v} \equiv a^{r(1+2c_v)} \equiv a \equiv t_i$ modulo $E_i^{\times n}$ by scalar centrality. This proves the criterion. No tameness assumption has been introduced. □

Example 3.2 (The diagonal rank-two check)**.** Take $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$. In the notation of Part I, Example 3.7, choose $c = 1$, so $c_v \equiv 2 \pmod 3$. Formula (5) becomes

$$A \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3, \qquad [\widetilde{t}, \widetilde{u}] = (t_1/t_2, u_1/u_2)_{3,F}.$$

Thus $\mathrm{diag}(t_1, t_2)$, when regular, is good exactly when $t_1/t_2 \in F^{\times 3}$. If a quadratic field E/F is embedded as a nonsplit maximal torus, the same theorem says that goodness is equivalent to $t \in F^\times E^{\times 3}$; the two tests are instances of the same formula, not separate conjectures.

Remark 3.3 (Why goodness is not enough for a regular-root formula)**.** The power of a regular element in a *split* torus need not be regular. For $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, the element $\mathrm{diag}(1, \zeta_3)$ is regular but not good: $\zeta_3 \notin F^{\times 3}$, since F does not contain ζ_9. Its cube is the identity. A character or orbital-integral formula with $\Delta(t^n)$ in the denominator must restrict to t^n regular (as in Zou's equation (3.2)), or use a separately proved germ extension. On a degree-r *field* torus, by contrast, if t is regular and $t^n \in F$, then Kummer theory gives $[F(t) : F] \mid n$; since $[F(t) : F] = r$ and $(r, n) = 1$, this is impossible. Because r is prime, $F(t^n) = F(t)$ and t^n is regular.

4 A new local power-map calculation for étale tori

The key local distinction is between a field centralizer (compact modulo scalars) and a product of fields (not compact modulo scalars). Both occur in the split *group*. The next proposition gives an exact count of the roots and the size of the image at wild and tame places, including the factors that a global trace identity must account for.

Lemma 4.1 (**[P]** Kummer injection for a product algebra)**.** *Let $E = \prod_{i=1}^k E_i$ be finite* étale *of dimension r over F, with $\mu_n \subset F$ and $(r, n) = 1$. Then*

$$F^\times / F^{\times n} \longrightarrow E^\times / E^{\times n} \tag{11}$$

is injective.

Proof. Suppose $a \in F^\times$ is an nth power in every E_i. The Kummer extension $F(a^{1/n})/F$, since $\mu_n \subset F$, is cyclic of degree $d \mid n$ and embeds in every E_i. Therefore $d \mid [E_i : F]$ for every i, so also $d \mid \sum_i [E_i : F] = r$. Coprimality gives $d = 1$ and $a \in F^{\times n}$. □

Proposition 4.2 (**[P]** Kernel, index, and Haar modulus)**.** *Under the hypotheses of Lemma 4.1, put* $A_E = E^\times/F^\times$ *and* $P_n([t]) = [t^n]$*. Then*

$$\ker(P_n) \simeq \mu_n(E)/\mu_n(F) \simeq \mu_n^k/\mu_n, \qquad |\ker(P_n)| = n^{k-1}, \tag{12}$$

$$[A_E : P_n(A_E)] = n^{2(k-1)}|n|_F^{-(r-1)}. \tag{13}$$

In particular, $P_n(A_E)$ *is open and closed. With* any *Haar measure* da *on* A_E *and its restriction to* $P_n(A_E)$*, for compactly supported integrable* q *on that image,*

$$\int_{A_E} q(a^n)\, da = n^{k-1}|n|_F^{-(r-1)} \int_{P_n(A_E)} q(b)\, db. \tag{14}$$

Proof. If $t^n = a \in F^\times$, then a is an nth power in every E_i; by Lemma 4.1, write $a = b^n$ with $b \in F^\times$. Thus each $t_i/b \in \mu_n(E_i) = \mu_n(F)$, giving (12), including its canonical description.

For a local field M containing μ_n the Kummer quotient has

$$|M^\times/M^{\times n}| = n^2|n|_M^{-1}. \tag{15}$$

Indeed the discrete valuation contributes n; the finite torsion unit factor containing μ_n contributes n; and the torsion-free p-adic unit factor contributes $|n|_M^{-1}$ (multiplication by n on its $\mathbb{Z}_p$-lattice). Alternatively, (15) follows from local duality. Since $F^\times/F^{\times n}$ injects into $E^\times/E^{\times n}$, the group A_E/A_E^n is its cokernel:

$$|A_E/A_E^n| = \frac{\prod_i n^2|n|_{E_i}^{-1}}{n^2|n|_F^{-1}} = n^{2(k-1)}|n|_F^{-(r-1)},$$

where $|n|_{E_i} = |n|_F^{[E_i:F]}$ for normalized absolute values on local fields. This proves (13).

For the Haar statement, write $K_0 = \ker(P_n)$, of order n^{k-1}. Choose a sufficiently small compact open subgroup $U \subset A_E$ consisting of principal-unit classes and having $U \cap K_0 = \{1\}$. The p-adic logarithm identifies U with a free $\mathbb{Z}_p$-module of rank $(r-1)[F : \mathbb{Q}_p]$: the factors $E_i^\times$ have combined principal-unit rank $r[F : \mathbb{Q}_p]$, and quotienting by the scalar principal units subtracts $[F : \mathbb{Q}_p]$. Multiplication by n maps U injectively into itself, with

$$[U : U^n] = p^{v_p(n)(r-1)[F:\mathbb{Q}_p]} = |n|_F^{-(r-1)}.$$

Since $P_n^{-1}(U^n) = K_0U$ and $K_0 \cap U = \{1\}$, its pushforward of Haar measure gives

$$\frac{\mathrm{vol}(P_n^{-1}(U^n))}{\mathrm{vol}(U^n)} = \frac{|K_0|\,\mathrm{vol}(U)}{\mathrm{vol}(U)/[U : U^n]} = n^{k-1}|n|_F^{-(r-1)}.$$

The pushforward is translation-invariant on the open image $P_n(A_E)$, so this ratio is its constant density on that image, proving (14). Notice that this argument requires neither a global quotient-index formula nor a choice of scale for Haar measure. □

Remark 4.3 (A measure-theoretic warning)**.** For a general open endomorphism $\varphi : A \to A$ of a locally compact abelian group with finite kernel, its pushforward Haar density need not equal $[A : \varphi A]/|\ker\varphi|$. For example, on $A = \mathbb{Z}$ with counting measure, $a \mapsto na$ has density 1 on $n\mathbb{Z}$, not n. The correct local computation here is $|\ker P_n|[U : U^n]$, where U is the compact open subgroup in the proof. For this specific A_E it happens to simplify to $[A_E : P_nA_E]/|\ker P_n|$; that numerical equality is not a general change-of-variables theorem.

Example 4.4 (Three local torus types)**.** If E/F is a field of degree r, then $k = 1$ and P_n is injective, with index and Haar modulus $|n|_F^{-(r-1)}$. If $E = F^r$ then $k = r$ and the kernel has order n^{r-1}. For $r = 2, n = 3$, $F = \mathbb{Q}_3(\zeta_3)$ and $|3|_F = 1/9$,

type	$\lvert\ker P_3\rvert$	$[A_E : A_E^3]$	factor in (14)
E/F quadratic	1	9	9
$E = F \times F$	3	81	27

These integers quantify why one cannot use the division-torus substitution formula unchanged on the full split group.

5 A global classification of good regular classes

The local theorem now applies to the actual torus at *every* place. Combining it with the Hasse principle of Part III removes the local-to-global ambiguity for power classes of a degree-r field, though not the independent measure and spectral questions.

Proposition 5.1 ([**E**] Almost-everywhere powers modulo scalars)**.** *Let E/K be a field extension of degree r with $(r, n) = 1$ and $\mu_n \subset K$. For every finite set Σ of places,*

$$E^\times/(K^\times E^{\times n}) \longrightarrow \prod_{v\notin\Sigma} \frac{(E \otimes_K K_v)^\times}{K_v^\times (E \otimes_K K_v)^{\times n}} \tag{16}$$

is injective. This is Part III, Proposition 11.3 [10].

Recalled proof. Let t be locally scalar times an nth power outside Σ. Choose $u, k \in \mathbb{Z}$ with $ru+nk = 1$ and put $a = N_{E/K}(t)^u \in K^\times$. If $t = a_v y_v^n$ in the local étale algebra, taking norms gives the identity

$$t/a = \big(a_v^k y_v / N_{E_v/K_v}(y_v)^u\big)^n.$$

Thus t/a is an nth power in every completion of E over all $v \notin \Sigma$. If its class in $E^\times/E^{\times n}$ were nontrivial, the nontrivial Kummer extension $E((t/a)^{1/n})/E$ would, by Chebotarev, admit infinitely many unramified places at which t/a is not an nth power. This contradicts the hypothesis. Hence $t \in K^\times E^{\times n}$. □

Theorem 5.2 ([**P**] Global good-regular criterion)**.** *Assume the covers and rational splittings of Part I. Let G denote $D^\times$ or GL_r, and let $\gamma \in G(K)$ be regular elliptic, so $E = K[\gamma]$ is a field of degree r. The following are equivalent:*

(a) the rational lift $\iota(\gamma)$ is good at every place;

(b) it is good at every place outside some finite set Σ;

(c) $\gamma \in K^\times E^{\times n}$.

The map $[s] \mapsto [s^n]$ is a bijection from the noncentral classes in $E^\times/K^\times$ onto the globally good regular classes belonging to E. Furthermore

$$W_{s^n} = W_s, \qquad W_s = \{\sigma \in \mathrm{Aut}(E/K) : \sigma(s) \in K^\times s\}. \tag{17}$$

Proof. At each finite v, the ordinary centralizer is $T_v = (E \otimes_K K_v)^\times$. If D_v splits, apply Theorem 3.1, transported through Part I, Proposition 2.6 (the split-place isomorphism of the specified covers). If D_v is division, use Part I, Proposition 3.5(a), whose commutator is the same by Part I, Proposition 3.1. At complex places the nth-power map on T_v is surjective. Thus at every place the local goodness condition is precisely $\gamma \in K_v^\times T_v^n$. Implications (c)⇒(a)⇒(b) follow, while (b)⇒(c) is Proposition 5.1.

The power map $E^\times/K^\times \to E^\times/K^\times$ is injective: if $s^n \in K^\times$, Kummer theory shows $[K(s) : K] \mid n$, while $[K(s) : K] \mid [E : K] = r$. Hence $s \in K^\times$. Because r is prime, every noncentral $s \in E$ generates E, and the same is true of s^n. This proves the bijection. Finally $\sigma(s^n)/s^n \in K^\times$ if and only if $(\sigma(s)/s)^n \in K^\times$, which is equivalent by injectivity to $\sigma(s)/s \in K^\times$. Equation (17) follows. □

Remark 5.3 (Precise consequence for the geometric side)**.** Part I, Proposition 6.1, uses the rational regular classes which are good everywhere, with coefficient

$$c(\gamma) = \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}. \tag{18}$$

Theorem 5.2 parametrizes its noncentral summation set by global $[s^n]$ and preserves W_γ. It does *not* replace the local root sums in Zou's character formula: the global root class may be unique even when a localized split torus has n^{k_v-1} roots modulo scalars.

6 The remaining wild local character problem

We distinguish character transfer from class-function transfer. The latter is a theorem of Part III; the former requires an *integral* statement not implied by unitarity.

6.1 Division-side normalization and discriminants

Let F be nonarchimedean, D/F a division algebra of prime degree r, $H = D^\times$, and $\widetilde{H}$ its specified cover. Set $Z = F^\times$ and $\widetilde{Z} = p^{-1}(Z)$, equipped with compatible Haar measures on $H/Z \simeq \widetilde{H}/\widetilde{Z}$. The Part III transform is

$$(L_D\Phi)(z\,t^*) = b_D\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}, \tag{19}$$

for regular t, $z \in \widetilde{Z}$; it vanishes at nongood regular lifts. Here

$$\Delta(t) = \left|\det\big(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t}\big)\right|_F^{1/2} = \prod_{i<j}\left|\frac{(\lambda_i - \lambda_j)^2}{\lambda_i\lambda_j}\right|_F^{1/2}. \tag{20}$$

This equals Part I's $|D(t)|^{1/2}|\,\mathrm{Nrd}(t)|_F^{-(r-1)/2}$. Since the power map on a field torus modulo scalars has index $|n|_F^{-(r-1)}$, Weyl integration gives b_D^2 equal to this index. The factor b_D is independent of *common* rescalings of the two identified quotient Haar measures. The ratio of discriminants extends at central elements to $|n|_F^{-r(r-1)/2}$ (Part III, Lemma 3.3), hence

$$(L_D\Theta_\rho)(1) = c_D \dim\rho, \qquad c_D = b_D|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2}. \tag{21}$$

The identity factor is pointwise and does not depend on Haar measures. Part III, Theorem 9.1, separately gives the adjoint orbital-integral/test-function transfer. Test functions transform by $\widetilde{\theta}^{-1}$, not by $\widetilde{\theta}$.

6.2 The integral matrix

Choose $\widetilde{\theta}$ first and put $\theta(a) = \widetilde{\theta}(a^*)$. Part III, Theorem 4.3 and Corollary 5.1, show that L_D is a unitary *isomorphism* of invariant L^2 spaces and that

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle, \tag{22}$$

with each row and column finite. The characters on either side are orthonormal bases in their fixed-central-character spaces.

Proposition 6.1 ([**E**] Integral-lattice criterion, Part III)**.** *The transform L_D permutes the irreducible characters with positive signs if and only if every $a_{\rho\tau}$ in (22) is an integer.*

Proof. An integral unit vector in ℓ^2 is a signed basis vector; unitarity gives injectivity and surjectivity because the inverse is the adjoint of a unitary map. The value at identity in (21) is positive and fixes the sign. This is Part III, Corollary 5.2, with its converse. □

Remark 6.2 (The actual obstruction)**.** For wild v, Part III does not prove $a_{\rho\tau} \in \mathbb{Z}$. The tame Adams argument works with a finite quotient Q and a central subgroup C, under the hypothesis $\gcd(n, |Q/C|) = 1$ (Part III, Lemma 7.5). At a wild place that hypothesis need not hold. For the simplest illustration take $Q = C_p$, $C = 1$, $n = p$: then the Adams operation sends every linear character to the trivial character. This does not prove a nonintegral coefficient for L_D; it only shows why that particular proof method fails.

6.3 A concrete first wild test

Consider again $r = 2, n = 3$, $F = \mathbb{Q}_3(\zeta_3)$. Since $[F : \mathbb{Q}_3] = 2$, one has $|3|_F = 3^{-2}$ and

$$I_F = 9, \qquad b_D = 3, \qquad c_D = 27. \tag{23}$$

For the trivial ρ on $D^\times$, compatible with a suitable choice of genuine central character, $L_D\Theta_\rho$ is

$$(L_D 1)(z\,t^*) = 3\,\widetilde{\theta}(z)\frac{|\operatorname{Nrd}(t)|_F}{|\operatorname{Trd}(t)^2 - \operatorname{Nrd}(t)|_F} \tag{24}$$

for noncentral t (and zero on nongood regular lifts). Indeed for the eigenvalues λ_1, λ_2 one has

$$\frac{\Delta(t)}{\Delta(t^3)} = \left|\frac{\lambda_1\lambda_2}{\lambda_1^2 + \lambda_1\lambda_2 + \lambda_2^2}\right|_F = \frac{|\operatorname{Nrd}(t)|_F}{|\operatorname{Trd}(t)^2 - \operatorname{Nrd}(t)|_F}.$$

Near a scalar $t = a$, the denominator is $|3a^2|_F$, so this ratio is 9, giving 27 at identity. Thus any genuine irreducible character equal to this function must have degree 27; *every* irreducible genuine character would have degree divisible by 27 if the entire transform were a bijection. Neither the existence of such a 27-dimensional representation nor the absence of smaller genuine representations has been established here.

A finite-level test would select a sufficiently small conjugation-invariant open subgroup on which the relevant representations are trivial (compatible with their central character), form the corresponding *twisted* finite quotient of H/Z, calculate its genuine or projective character table, and evaluate the finite inner products (22). A plain quotient $\widetilde{H}/(\widetilde{Z}U)$ cannot be used by itself as a "genuine character table": dividing out by $\widetilde{Z}$ destroys the prescribed genuine central action. The central character must be retained as a multiplier in the finite twisted group algebra. This computation has not been carried out in this draft.

7 Connection with Zou's corrected character relation

For the Kazhdan–Patterson cover of the *split group*, Zou [3, §3] revisits the Flicker–Kazhdan metaplectic correspondence and records a corrected scalar in the character relation. This is an essential input, but its objects are $\widetilde{\mathrm{GL}_r(F)}$ and ordinary $\mathrm{GL}_r(F)$, not directly $\widetilde{H}$ and H at a division place.

Proposition 7.1 (**[E]** Zou's root-sum constant in our case)**.** *Write $d_F(m) = m|m|_F^{-1/2}$ and $d_r = \gcd(n, 2rc_v + r - 1)$. Under (1), $d_r = n$. In Zou's equation (3.2), the coefficient preceding the root sum in rank r is*

$$\frac{1}{d_F(d_r)|n|_F^{r/2}} = \frac{1}{n|n|_F^{(r-1)/2}} = \frac{b_D}{n}. \tag{25}$$

For regular elliptic x with degree-r field centralizer, x^n is automatically regular by Remark 3.3. The equation $t^n = x^n$ then has exactly n solutions $t = \zeta x$, $\zeta \in \mu_n$. If the linear character π has central character ω_π trivial on μ_n, these summands agree and the effective coefficient is b_D. This restriction on ω_π is essential: without it, the scalar sum is $\sum_{\zeta\in\mu_n} \omega_\pi(\zeta) = 0$, so equation (3.2) instead forces the covered character to vanish at x^*. For a genuine lift pair the compatibility $\omega_\pi(a) = \omega_{\widetilde{\pi}}(a^*)$ and $\zeta^* = 1$ imply $\omega_\pi|_{\mu_n} = 1$ automatically.

Proof. The equation $d_r = n$ is immediate from scalar centrality; (25) follows by substitution. Regularity of x^n follows from the field case of Remark 3.3. Any solution t centralizes x^n, so $t \in F[x]^\times$. In the field $F[x]$, $(t/x)^n = 1$, and $\mu_n(F[x]) = \mu_n(F)$. Thus $t = \zeta x$. The Weyl discriminant is invariant under central scalar multiplication, the ordinary character takes the same value because it is trivial on μ_n, and $(\zeta x)^* = x^*$ by the explicit odd-n scalar star computation in Section 2.2. These are precisely the equal terms in Zou's (3.2). □

Remark 7.2 (What is and is not composed)**.** At the ramified places one may compare the *desired* $\overline{G_{D_v}} \leftrightarrow \overline{G}_v$ character relation of Parts I–II with (a) the classical linear Jacquet–Langlands elliptic character identity and (b) Zou's formula. The common coefficient b_D and the familiar sign $(-1)^{r-1}$ are consistent with this comparison. Nevertheless *the existence of an irreducible genuine image on the covered division side is the missing conclusion*; combining two known formulas cannot be used to assume that missing representation exists. For non-elliptic split centralizers, retain Zou's entire root sum, including the central-character factors.

8 Global geometric comparison: proved input and new obligations

There are two comparisons to keep distinct:

$$\begin{aligned} \overline{G_D}(\mathbb{A}_K) &\longleftrightarrow \overline{G}(\mathbb{A}_K) \quad \text{(Parts I–II, under their hypotheses)}, \\ D^\times(\mathbb{A}_K) &\dashrightarrow \overline{G_D}(\mathbb{A}_K) \quad \text{(proposed additional comparison)}. \end{aligned} \tag{26}$$

The first has a proved simple trace-formula comparison with regular-support or later matching hypotheses (Part I, Theorem 6.6; Part II, Theorem 6.2). The second is not a formal consequence of the first or of the unitarity of L_D.

8.1 Geometric coefficients and the product formula

For a regular elliptic rational γ, with $E = K[\gamma]$, Part I identifies the coefficient

$$c(\gamma) = \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}.$$

The same torus measure dt must be used whenever this term is compared with the corresponding linear inner-form orbital integral. Part I uses the normalized orbital integral $J_v(\gamma, f) = \Delta_v(\gamma)O_v(\gamma, f)$. For a rational regular element the product formula implies

$$\prod_v \Delta_v(\gamma) = 1. \tag{27}$$

This is because each discriminant is the normalized absolute value of a nonzero element of K (or its square root), including the determinant normalization in (20). The local *ratio* of discriminants $\Delta_v(t)/\Delta_v(t^n)$ must be kept until local transfer and the global product are both defined.

At $v \in S$ Part III, Theorem 9.1, gives the exact local normalized orbital-integral identity

$$J_{\widetilde{H}_v}(z\, t^*, \mathcal{O}_{D,v} f_v) = b_{D,v}\, \widetilde{\theta}_v(z)^{-1} J_{H_v}(t, f_v), \tag{28}$$

using the same quotient measures on the two copies of H_v/Z_v. At $v \notin S$, $H_v \cong \mathrm{GL}_r(K_v)$ and the appropriate root-sum comparison is not (28); it must use Flicker–Kazhdan–Zou, and it can involve the kernel $\mu_n^{k_v}/\mu_n$ of Proposition 4.2. In particular the local factors $b_{D,v}$ for $v \in S$ *cannot* simply be cancelled using $\prod_v |n|_v = 1$ without the split-side contributions.

8.2 Haar ledger

For $v \in S$, Part II, Definition 2.1, fixes

$$\mathrm{vol}(D_v^\times / F_v^\times; dg_{D,v}) = d(\mathrm{St}; dg_{\mathrm{GL},v})^{-1}, \tag{29}$$

where $d(\mathrm{St}; dg_{\mathrm{GL},v})$ is the formal degree relative to the chosen $\mathrm{GL}_r(F_v)/F_v^\times$ measure. The probability measure used in Part III is $d(\mathrm{St}; dg_{\mathrm{GL},v})\, dg_{D,v}$. When *both* sides of (28) are scaled by the same measure, $b_{D,v}$ is unchanged. Different measures on the GL side must still be reconciled with (29).

Item	Required statement or calculation	Status
Ambient cocycle and common center	Part I, Lemma 5.1 and Prop. 5.5; restriction (3)	[E]
Regular rational good locus	Theorem 5.2	[P]
Global rational/stable classes	Hilbert 90 and Kottwitz's framework; local embeddability into D	[E]
Global torus coefficient	Same dt in (18), including $\lvert W_\gamma \rvert$	[E]
Local $v \in S$ orbital transfer	Part III, Theorem 9.1, equation (28)	[E]
Split-place $v \notin S$ invariance	Lemma 8.2; [2, §6]	[P]
Split-place pointwise transfer and λ_v	Proposition 8.3; unproved existence	[C]
Unramified unit elements	[2, §12] in its range; splitting compatibility must be checked	[E]/[O]
Nonregular powers and singular germs	Remark 3.3; germ compatibility or support needed	[C]
Identity and singular terms	Part I, Prop. 6.1; regular support eliminates them, otherwise new comparison required	[E]/[C]
Spectral expansion	Part I–II packet formula with restrictions; new factor matching	[E]/[C]

Here N_r is the least common multiple of the *composite* positive integers at most r (the convention in [FK]). The classical unramified matching used in [FK, §12] is established under $(n, N_r) = 1$; outside that range one must supply an alternative theorem rather than erase the assumption.

8.3 Split-place invariance and a geometric comparison criterion

At a place $v \notin S$ the algebra D_v is split, but a rational elliptic field $K[s]$ may decompose locally as a product of fields. By Proposition 4.2, its local nth-root fibre modulo scalars can contain n^{k_v-1} points. The global class $[s]$ supplied by Theorem 5.2 is nevertheless unique. Therefore a local sum over *all* roots cannot be substituted for the orbital integral of this one rational representative without an additional compatibility statement.

Definition 8.1 (FK-compatible ordinary functions)**.** Fix the local central characters $\theta_v, \widetilde{\theta}_v$ with $\theta_v(a) = \widetilde{\theta}_v(a^*)$. A normalized ordinary test function f_v is called *FK-compatible on the regular-power locus* if, whenever $x, x' \in H_v$ have x^n, x'^n regular and their star images satisfy $(x')^* = zx^*$ for a central lift $z \in \widetilde{Z}_v$ (after conjugating representatives), one has

$$J_{H_v}(x', f_v) = \widetilde{\theta}_v(z)^{-1} J_{H_v}(x, f_v). \tag{30}$$

In particular $J_{H_v}(\xi x, f_v) = J_{H_v}(x, f_v)$ whenever $\xi \in \mu_n$ and both powers are regular. Equation (30) is the invariant-function condition of [2, §6], expressed with the present inverse central transformation convention.

Lemma 8.2 ([**P**] Necessary split-place invariance)**.** *Suppose a genuine $\widetilde{f}_v$ and a fixed nonzero constant λ_v satisfy*

$$J_{\widetilde{H}_v}(x^*, \widetilde{f}_v) = \lambda_v J_{H_v}(x, f_v) \tag{31}$$

for all x in a conjugation-invariant region where x^n is regular, and suppose the corresponding identity holds also at central translates of star images. Then f_v obeys (30) on that region.

Proof. If $(x')^* = zx^*$, conjugation invariance and the inverse central transformation law of the genuine test function imply $J_{\widetilde{H}_v}((x')^*, \widetilde{f}_v) = \widetilde{\theta}_v(z)^{-1} J_{\widetilde{H}_v}(x^*, \widetilde{f}_v)$. Apply (31) twice and divide by the fixed nonzero λ_v. In particular, for $x' = \xi x$ with $\xi \in \mu_n$, one has $(x')^* = x^*$, yielding constancy on those scalar-root fibres. Thus FK-compatibility is a *necessary* condition, not an asserted construction of $\widetilde{f}_v$. □

A regularity and germ obligation. Equation (31) is imposed only where x^n is regular. On a split torus, x can be regular while x^n is singular (Remark 3.3); the power map is a local analytic isomorphism on the torus, but normalized orbital integrals of a function at images approaching a singular element may have nonconstant Shalika-germ behavior. A full transfer construction must either arrange support away from these root-collision loci or prove the matching germ conditions there. No universal vanishing of the germ coefficients is claimed here.

Proposition 8.3 ([**P**] Conditional geometric comparison with explicit split-place data)**.** *Fix the rational splittings and quotient Haar measures as in Parts I–III. Choose factorizable ordinary and genuine test functions and a regular elliptic support condition at an auxiliary place so that the relevant simple trace formulas have no identity/singular contribution. Suppose:*

(i) For each $v \notin S$, the ordinary function f_v is FK-compatible in the sense of Definition 8.1, and a genuine test function $\widetilde{f}_v$ satisfies (31) for every regular rational local component s_v of a globally good class $[s^n]$, with a constant $\lambda_v \in \mathbb{C}^\times$ independent of s_v. The same central-translate identity holds, and $\lambda_v = 1$ for almost all v. This includes the complex archimedean places.

(ii) For each $v \in S$, use Part III, Theorem 9.1, with $\widetilde{f}_v = \mathcal{O}_{D,v} f_v$, the coefficient $b_{D,v} = |n|_v^{-(r-1)/2}$ and exactly the same quotient measures on H_v/Z_v and $\widetilde{H}_v/\widetilde{Z}_v$. The test functions have the prescribed reciprocal central transformation laws.

(iii) The local constants satisfy the specific *product condition*

$$\Big(\prod_{v\notin S}\lambda_v\Big)\Big(\prod_{v\in S} b_{D,v}\Big)=1. \tag{32}$$

The same torus Haar measures, rational splittings, normalizer orders, and central phases are used on the two sides; the simple trace-formula expansions converge in the stated test range.

Then the regular elliptic geometric expansions for $D^\times$ and its specified cover agree, and consequently so do their simple spectral trace distributions in that range.

Proof. Theorem 5.2 parametrizes the contributing globally good noncentral rational classes by $[s^n]$, with a unique global root class $[s]$ and the same normalizer orders. FK-compatibility removes ambiguity from the multiple *local* roots without introducing extraneous rational summands. At a place outside S, hypothesis (i) supplies $\lambda_v J_{H_v}(s_v,f_v)$, with the fixed central phase; at a place in S, hypothesis (ii) and (28) supply $b_{D,v}J_{H_v}(s_v,f_v)$. The torus coefficients (18) are the same and the product of the Weyl discriminants equals 1 by (27). The scalar left over in each rational geometric term is the left side of (32), hence is 1. The support condition eliminates the singular terms, and convergence justifies summing. Equality of the corresponding spectral distributions follows from the two trace formulas. The *implication* is proved; the existence of the split-place functions, the constants and the product identity in (i)–(iii) are additional research problems, not consequences of this proposition. □

8.4 Why Li, Kottwitz, and DKV do not close the gap

The linear Jacquet–Langlands theorem of Deligne–Kazhdan–Vignéras [1] controls the ordinary $D^\times$–GL_r relation, including the elliptic character sign, but it does not automatically manufacture a genuine representation of $\overline{G_D}_v$. Kottwitz's analysis of stable conjugacy and relative trace coefficients [5] provides the correct arithmetic setting; for the induced tori $\mathrm{Res}_{E/K}\mathbb{G}_m$ one has $H^1(K,T)=1$ by Shapiro's lemma and Hilbert 90, and embeddings $E\hookrightarrow D$ are controlled by Brauer splitting. Wen–Wei Li proves an invariant trace formula for adelic covers under an invariant trace Paley–Wiener hypothesis for Levi subgroups at *real* archimedean places [7]. Here K is totally imaginary, so that real-place hypothesis is vacuous. At the complex places the specified covers split (Part I, Proposition 2.3 and its archimedean remark), and the linear invariant Paley–Wiener theorem of Clozel–Delorme [6] supplies the relevant archimedean background. Li also discusses known real metaplectic GL cases, but his general result does not assert the particular *equality* of ordinary-$D^\times$ and covered-$D^\times$ traces required in (26). The simpler already-proved trace formulas of Parts I–II remain the appropriate initial comparison.

9 Spectral separation and a research program

9.1 The actual spectral distribution identity

Let $H=D^\times$ and $\widetilde{H}=\overline{G_D}$ globally, with compatible central characters $\theta,\widetilde{\theta}$. The quotient $H(K)Z_H(\mathbb{A}_K)\backslash H(\mathbb{A}_K)$ is compact because D is a division algebra over K; its covered analogue is compact modulo the specified central preimage as well. Thus the fixed-central-character right regular representations have discrete Hilbert decompositions. For factorizable test functions with

the inverse central transformation law, define

$$I_H(f) = \sum_{\rho \in \mathcal{A}(H,\theta)} m_H(\rho) \prod_v \operatorname{tr} \rho_v(f_v), \tag{33}$$

$$I_{\widetilde{H}}(\widetilde{f}) = \sum_{\widetilde{\rho} \in \mathcal{A}(\widetilde{H},\widetilde{\theta})} m_{\widetilde{H}}(\widetilde{\rho}) \prod_v \operatorname{tr} \widetilde{\rho}_v(\widetilde{f}_v). \tag{34}$$

Here $\mathcal{A}$ denotes the relevant automorphic spectrum and m is its actual spectral multiplicity, not an assumed 1. The expressions are interpreted on trace-class test functions satisfying the simple trace-formula hypotheses; a bare unrestricted infinite sum is not justified by notation alone. Under the assumptions of Proposition 8.3, the equality of the geometric expansions supplies $I_H(f) = I_{\widetilde{H}}(\widetilde{f})$. It does not identify summands. This is a conditional identity of *distributions*, with both the multiplicities and local trace factors retained.

9.2 Individual local characters and integrality

An equality of global spectral sums is a statement about *distributions*. To isolate a single local irreducible at a wild place, one needs an argument that the relevant coefficients are integers (or multiplicities which can be shown to isolate a single constituent). Linear independence alone can separate finite lists of characters; it does not supply the required integrality from a unitary Fourier transform.

Remark 9.1 (What an integer spectral coefficient would prove)**.** If a justified global-to-local argument supplies $a_{\rho\tau} \in \mathbb{Z}$ for every τ in the row of a fixed ρ, Proposition 6.1 (Part III, Corollary 5.2) forces $L_{D,v}\Theta_\rho$ to be one irreducible genuine character with positive sign. If integrality holds for all rows, unitarity implies a bijection. This is an application of an already proved lattice criterion, not a second conditional theorem; the missing step is precisely the integrality hypothesis at a wild place.

Separating the remaining tasks. First, the local wild problem can be tested using the explicit rank-two example in Section 6.3; it requires an actual twisted finite character table and a proof that the chosen finite level captures the desired row. Second, at split places the root sum in Zou's (3.2) should be written against the kernel and index in Proposition 4.2, including the chosen Haar measures and central phases. Third, one needs matching unramified test functions under the precise $(n, N_r) = 1$ or an improved hypothesis, and a complete global coefficient identity as in Proposition 8.3. Finally, one must justify globalization and spectral separation for representations of the covered inner form, and remove the auxiliary supercuspidal and non-cancellation hypotheses of Parts I–II. These are mathematical work packages, not restatements of an already-proved theorem.

10 Selected glossary

Good element A regular lift commuting with the full preimage of its ordinary centralizer; genuine invariant characters vanish on the nongood regular locus.

Scalar-central cover
A cover whose entire inverse image of $F^\times I_r$ is central; here forced by $n \mid r-1+2rc_v$.

Power-map kernel
$\ker(P_n)$ on $T(F)/F^\times$, distinct from the image index. The kernel can be nontrivial on product tori even when the corresponding field-torus map is injective.

Star map $h^* = \widetilde{h}^{\,n}$, independent of lift; multiplication on tori uses n odd.

Weyl discriminant
The factor $\Delta = |\det(1 - \mathrm{Ad}(t)|\mathfrak{g}/\mathfrak{t})|^{1/2}$; it is not the unnormalized characteristic-polynomial discriminant.

Twisted finite quotient
A finite quotient of H/Z with the central character retained as a multiplier, used to compute projective/genuine irreducible characters.

Packet identity
An identity of traces over collections of automorphic representations; it does not assert a bijection between individual local irreducibles.

Formal degree The Plancherel normalization of a square-integrable representation relative to a specified quotient Haar measure.

Root-sum relation
Zou's equation (3.2), which sums over all nth roots satisfying the stated regularity condition rather than choosing one arbitrarily.

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